

August 22, 2026

Prof. Jin Xuan and Prof. Jinfeng Liu
Editors-in-Chief
Digital Chemical Engineering
Elsevier

Dear Professors Xuan and Liu,

Please consider my original research article, “Optimization of a Recycling N-Stage Activated Sludge System with a Clarifier using A Physically Constrained Surrogate,” for publication in *Digital Chemical Engineering*.

The manuscript develops a closed-loop design-and-operation framework for a prescribed N -stage train of continuous stirred-tank reactors connected to a secondary Clarifier. The decisions include nominal hydraulic retention time, zonal aeration, mixed-liquor recycle, return activated sludge, and waste activated sludge. One system-level second-order surrogate jointly predicts the mixer, every reactor, Clarifier outlet component mass flows, and the ten-layer Clarifier solids profile.

The principal methodological contribution is a unique scaled- L_2 lower-level quadratic program for the interconnected bioreactor–Clarifier network. It enforces exact recycle mixing, interstage stoichiometric invariants, componentwise Clarifier mass balance, soluble pass-through, particulate clarification and densification directions, layer-inventory consistency, and non-negativity. RAS densification is represented as particulate recovery into a smaller underflow rather than artificial solids generation. The upper level evaluates clarified effluent quality together with volume, aeration, recycle-pumping, and wasted-solids terms and checks SRT and Clarifier-capacity constraints without altering the surrogate response to manufacture compliance.

The study builds on the component-space and invariant ideas in the published work of Selerio Jr. (2026), *An interpretable statistical surrogate of activated sludge systems that preserves mass conservation and component non-negativity*, *Digital Chemical Engineering*, 20, 100329. The present model uses dimensionless unique quadratic monomials and one identifiable convex, physics-penalized regression, followed by a candidate-dependent network projection. The original single-reactor coefficients and dataset are not reused.

Training and validation use a coupled ASM2d-TSN–multilayer-Clarifier model. Predictive fidelity is evaluated on held-out states and at surrogate-selected decisions. Decision quality is assessed separately through an independent mechanistic search and reported as a gap to the best finite-budget incumbent found under the same bounds. The formulation and computational record distinguish a fast physically admissible prediction from evidence about the associated engineering decision.

The work is aligned with the journal’s scope in digital process engineering, mathematical programming, physically constrained machine learning, and computational wastewater-process operation. The manuscript is original, has not been published previously, and is not under consideration elsewhere. I have approved its submission and declare no competing interests. Thank you for your

consideration.

Sincerely,

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